



ADNAN KHAN

Operations Research | Stochastic Optimization | Supply Chain Analytics

✉ adnan.khan@student-cs.fr ☎ 0758454551 🔗 linkedin.com/in/adnankhan121 📍 Paris, France

🎓 EDUCATION

Operations Research & Risk Analytics (M2), CentraleSupélec 09/2025 – 10/2026 | Paris, France
• Grade: 14/20 (Approx.) | Institutional Scholarship Holder

Industrial Engineering (B.Eng), University of Dhaka 04/2020 – 12/2024 | Dhaka, Bangladesh
• CGPA: 3.50/4.00 | 87th percentile in the department.

🧠 TECHNICAL SKILL

Solvers and Modelling — Gurobi, Pyomo, Excel Solver

Programming and Data — C/C++, Python (NumPy, Pandas, SciPy, Matplotlib), SQL, Power BI

Optimization — MILP, Duality, Sensitivity Analysis, SAA, Benders Decomposition, Vehicle Routing, Network Design

📁 EXPERIENCE

Institut du Cerveau (Paris Brain Institute), Intern, Modelling and Data Analysis 06/2026 – Present | Paris, France
Gaussian HMMs and analysis pipelines in Python; maximum likelihood estimation and Viterbi decoding on experimental time series to extract latent states, transition probabilities and dwell times.

Akij Food and Beverage Limited (AFBL), Industrial Engineering Intern 05/2024 – 07/2024 | Bangladesh
Time and motion studies across 4 layouts to locate throughput bottlenecks; redesigned production layout using LEAN.

Kabir Steel Re-Rolling Mills (KSRM), Production Intern 06/2023 – 08/2023 | Chattogram, Bangladesh
Cut production defects 11% over a 27-day tracked cycle; Pareto and 5-Why root cause analysis to isolate dominant failures.

📁 PROJECT

Food Bank Inventory Management under Stochastic Demand 01/2026 – 02/2026
Two-Stage SAA, Network Optimization, Equity Constraints, Python

- Two-stage stochastic over weekly orders and daily transportation under correlated demand with equity constraint.
- Extensive SAA on 260 weeks: 93.8% fill rate, equity gap ~2% limit; quantile based dynamic policy validated out-of-sample.

Optimization of Maintenance Planning for Offshore Wind Turbines 11/2025 – 12/2025
Python (Pyomo, Gurobi) | MILP, Maintenance Scheduling, Reliability

- MILP scheduling preventive and corrective maintenance under crew, vessel-travel and maintenance-window constraints.
- Discrete health states; minimized lifecycle cost over maintenance, transport, outage penalties; sensitivity tested.

Stochastic Optimization Projects 03/2026 – 04/2026
Stochastic Programming, Benders Decomposition, CVaR, Markov Decision Processes

- Capacity planning: two-stage stochastic LP with recourse, solved by SAA extensive form and Benders L-shaped decomposition; extended to risk-averse CVaR.
- Machine replacement: infinite-horizon MDP over ten states solved by value iteration on the Bellman equation; sensitivity on cost, discount factor and transition probabilities.

📄 THESIS

Integrated Vehicle Routing, Order Assignment and Optimization for Resilient Omnichannel Distribution Networks, CentraleSupélec | Supervisor: Prof. Jakob Puchinger | MILP, Vehicle Routing, Python (Pyomo, Gurobi)

- MILP integrating vehicle routing and order assignment across an omnichannel network, extending the capacitated VRP with channel-specific constraints.
- Benchmarked integrated against sequential baseline; quantified network resilience by re-solving under disruption.

🏆 COMPETITION

- TOP 10 teams in the BIHRM Supply Chain Contest, achieved 75% merit-based scholarship in SC professional program.
- Regionally selected and competed at the national level in the Bangladesh Math Olympiad (2016).

🌐 LANGUAGE

English - C1 Level (British Council)

French - B1 Level (FLE)

Bengali - Native Level